Mycoplasma Encephalitis (MONDO:0019377): A Comprehensive Disease Characteristics Report

Summary

Mycoplasma encephalitis (MPE) is an acute, predominantly pediatric inflammation of the brain triggered by infection with *Mycoplasma pneumoniae* (Mp; NCBI:txid2104), the cell-wall–deficient, atypical respiratory bacterium that is one of the most commonly identified infectious causes of childhood encephalitis. Across large referral and prospective cohorts, *M. pneumoniae* accounts for roughly **5–6% of confirmed childhood encephalitis cases** and is the single most frequently identified bacterial pathogen in several encephalitis surveillance projects. It is not a Mendelian, single-gene disorder; it is an infectious/immune-mediated condition (Infectious Disease category), and no causal human gene or established animal model of the extrapulmonary (neurologic) disease exists. Host susceptibility is polygenic/multifactorial, with candidate immune-gene loci modulating *severity of M. pneumoniae respiratory disease* rather than causing encephalitis specifically.

Mechanistically, MPE injures the central nervous system by at least **three overlapping routes**: (1) an **early-onset, direct/CNS-invasive and toxin-mediated** pattern in which the organism reaches the CSF and the CARDS (Community-Acquired Respiratory Distress Syndrome) toxin plus TLR–MyD88–IL-8 innate neuroinflammation drive damage; (2) a **late-onset, post-infectious autoimmune** pattern driven by molecular mimicry, producing anti-galactocerebroside, antiganglioside (anti-GM1), and anti-NMDA-receptor antibodies that cause demyelination/ADEM and other syndromes; and (3) a **vaso-occlusive/thrombotic** route producing cerebral (often watershed) infarction through focal vasculitis, hypercoagulability, and hypoperfusion. These branches are distinguished clinically by prodrome length and by whether the organism is detectable in CSF.

Clinically, MPE presents with altered consciousness, seizures (frequently progressing to refractory status epilepticus in severe cases), headache, fever, and near-universal preceding/concurrent respiratory illness. Diagnosis relies chiefly on **serology** (CSF *M. pneumoniae* PCR is positive in only ~2% of cases). Treatment combines **CNS-penetrating anti-mycoplasmal**

antibiotics (macrolides, tetracyclines, fluoroquinolones — β-lactams are ineffective because Mp lacks a cell wall) with **immunomodulation** (corticosteroids, IVIG, plasma exchange). Prognosis is generally favorable with low mortality (≈2.6% in adult meningoencephalitis), but **neurologic sequelae occur in roughly 25–30%**, and MPE complicated by refractory status epilepticus carries substantial morbidity and mortality.

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1. Disease Information

Overview. Mycoplasma encephalitis is acute encephalitis (inflammation of the brain parenchyma, often with meningeal involvement → meningoencephalitis) occurring during or shortly after *M. pneumoniae* infection. Neurologic disease is the **most common extrapulmonary complication** of *M. pneumoniae*, with a total incidence of about **0.1% of infections**, and encephalitis is the most common neurologic syndrome within that spectrum (PMID: 41697567; PMID: 26084846).

Key identifiers. - **MONDO:** MONDO:0019377 (Mycoplasma encephalitis) - **MeSH:** *Encephalitis* (D004660) with *Pneumonia, Mycoplasma* (D011019) / *Mycoplasma pneumoniae* (D009976); no dedicated MeSH descriptor for the encephalitis subtype - **ICD-10:** Best mapped via *G05.3* (Encephalitis in other diseases classified elsewhere) plus *J15.7* / *A49.3* (*M. pneumoniae* infection); ICD-11: *1D01* (encephalitis) with an infectious-agent extension code - **OMIM/ Orphanet:** Not applicable as a Mendelian entry; not a rare genetic disease with an OMIM phenotype number - **NCBI Taxonomy (causal agent):** *Mycoplasma pneumoniae* NCBI:txid2104

Synonyms / alternative names: *M. pneumoniae* encephalitis, *Mycoplasma pneumoniae*-associated encephalitis, mycoplasmal meningoencephalitis, *M. pneumoniae* meningoencephalitis, *M. pneumoniae*-associated CNS disease.

Information source type. This report is derived from **aggregated disease-level resources** — published case series, prospective cohorts (California Encephalitis Project, Australian Childhood Encephalitis study, Toronto SickKids series), systematic reviews, and mechanistic/ microbiology studies — rather than individual EHR data.

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2. Etiology

Primary cause — infectious. MPE is caused by infection with *M. pneumoniae*, a small, self-replicating, cell-wall–deficient bacterium that colonizes the respiratory epithelium via specialized attachment organelles ("gliding" colonization) (PMID: 42211665). It is **not genetic** in origin.

Genetic risk factors (host susceptibility to Mp disease, not MPE-specific). No germline variant causes MPE. Candidate-gene studies of *M. pneumoniae* pneumonia (MPP) susceptibility in Chinese children have identified associated loci: **ACE rs4340, IL-6 rs1800795, and NOS3 rs1799983** (n=415 cases/300 controls; PMID: 28403117); functional eQTL-SNPs including **rs8032531** (PMID: 35836993); and **IL2 rs6534349 / IL4 rs2227284** linked to Mp infection and asthma risk (PMID: 25747600). These modulate respiratory-disease susceptibility/severity and are best regarded as polygenic modifiers, not causes of encephalitis.

Environmental risk factors. - **Age:** pediatric predominance (median ~7 years for MPE) (PMID: 41935250) - **Season:** winter/epidemic peaks; Mp epidemics recur every **3–7 years**, with a post-COVID-pandemic resurgence in 2023–2024 (PMID: 42560049; PMID: 42283501) - **Sex:** modest male predominance in pediatric MPE (43M:28F in one cohort) - **Exposure:** crowded settings (schools, households) favor transmission

Protective factors. No validated genetic protective variants for MPE are established. At the level of *respiratory* disease progression, early β-lactam therapy and EBV coinfection were paradoxically inversely associated with fulminant Mp pneumonia in one case–control study (PMID: 41772495) — a respiratory, not neurologic, endpoint. Prompt appropriate anti-mycoplasmal therapy is the principal modifiable protective factor.

Gene–environment interactions. Intrafamilial cases infected by **identical A2063G macrolide-resistant strains** showed markedly different disease severity, indicating that **host immune maturity, rather than organism virulence, is the dominant determinant of severity** (PMID: 19898978). This is the clearest documented gene/host–environment interaction: identical pathogen exposure, divergent host outcome.

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3. Phenotypes

MPE produces a broad neurologic phenotype superimposed on a systemic inflammatory/respiratory illness.

Phenotype	Type	Frequency	HPO suggestion
Impaired/altered consciousness	Clinical sign	90% (pediatric) / 67% (mixed)	HP:0011446 (abnormal consciousness)
Headache	Symptom	87.4%	HP:0002315
Fever	Symptom	96.5%	HP:0001945
Respiratory involvement	Sign	94.3%	HP:0002086
Seizures	Clinical sign	~28–39%; status epilepticus in severe cases	HP:0001250
Movement disorders	Clinical sign	~38% in basal-ganglia subset	HP:0100022
Behavioral/psychiatric change	Behavioral	common (esp. anti-NMDAR overlap)	HP:0000708
Ataxia / cerebellar signs	Clinical sign	~10% of neuro cases	HP:0001251
Elevated CRP	Laboratory	90.8%	HP:0011227
CSF pleocytosis	Laboratory	~66.7% (basal-ganglia subset)	HP:0012229

Onset: acute to subacute, typically within days to <2 weeks of the respiratory prodrome. **Severity:** variable — from mild, reversible (MERS-type splenial lesion) to severe with refractory status epilepticus. **Progression:** usually monophasic/acute; episodic seizures may occur acutely; most cases are self-limited with treatment.

Quality-of-life impact. In survivors with sequelae, impacts include epilepsy, motor deficits, cognitive/behavioral impairment, and (in a documented hypothalamic case) secondary hypersomnia/narcolepsy-like symptoms with decreased CSF hypocretin-1 (PMID: 22419016). No disease-specific EQ-5D/SF-36 data exist for MPE.

Quantitative phenotype data: pediatric MPE (n=87) — consciousness disturbance 90%, headache 87.4%, fever 96.5%, respiratory involvement 94.3%, multisystem involvement 98.2%, elevated CRP 90.8% (PMID: 37417436). Adult meningoencephalitis (n=39) — MRI abnormalities 78.9%, leptomeningeal enhancement 55.3%, EEG abnormalities 35.9% (PMID: 42624009).

4. Genetic / Molecular Information

Causal genes: None. MPE has no Mendelian cause, no causal human gene, and no pathogenic germline variant. It is not listed in ClinVar/OMIM as a monogenic disorder.

Modifier / susceptibility genes (host, respiratory-disease level): ACE (rs4340), IL6 (rs1800795), NOS3 (rs1799983), IL2 (rs6534349), IL4 (rs2227284), and eQTL-SNP rs8032531 — all associated with Mp pneumonia susceptibility/severity, not MPE per se (PMID: 28403117; PMID: 35836993; PMID: 25747600).

Pathogen genetics (clinically actionable): Macrolide resistance in *M. pneumoniae* is driven predominantly by the **23S rRNA A2063G point mutation**, with high prevalence in China (PMID: 42568601; PMID: 42757852). The 2023–2024 resurgence coincided with clonal replacement by a **P1-1/ST3 lineage** carrying fixed A2063G macrolide resistance (PMID: 41586312).

Epigenetic / chromosomal information: Not applicable — no human epigenetic or chromosomal abnormality is associated with MPE.

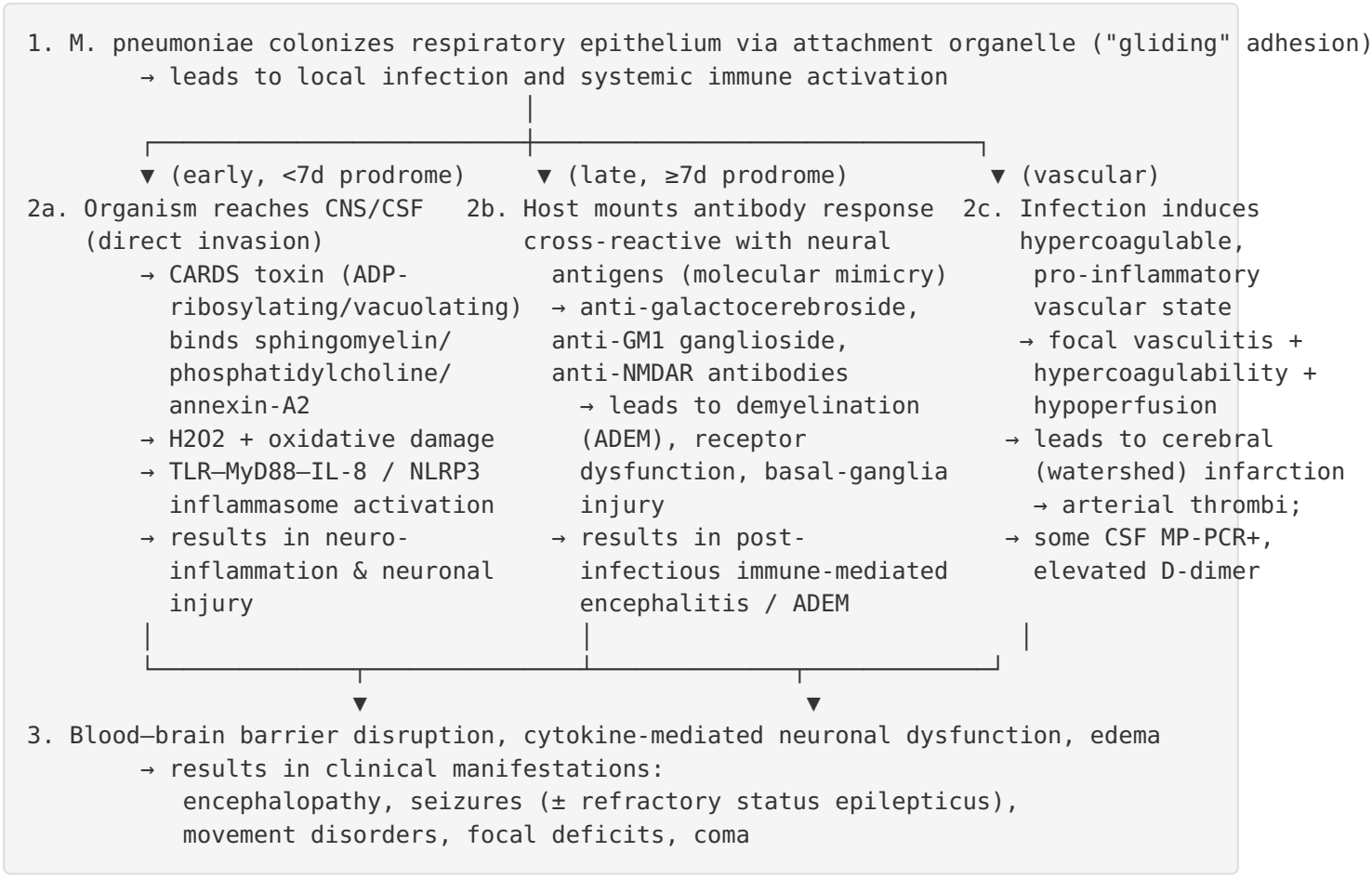
5. Environmental Information

Infectious agent: *Mycoplasma pneumoniae* (NCBI:txid2104) — the sole and necessary cause. It is a cell-wall–deficient bacterium (class Mollicutes) transmitted person-to-person via respiratory droplets.

Environmental/lifestyle factors: Seasonality (winter peaks), epidemic cycling every 3–7 years, and crowded living/schooling conditions increase exposure. No chemical toxin, radiation, or occupational exposure is implicated. CTD-type toxicant associations are not relevant to this infectious disease.

6. Mechanism / Pathophysiology

Ordered causal chain



Detail by category

Molecular pathways / innate immunity. The **TLR → MyD88 → IL-8** axis and the **NLRP3 inflammasome** are central innate drivers. CARDS toxin, MyD88, and IL-8 expression are positively correlated with severe Mp pneumonia and disease severity (PMID: 42572866); Mp activates the NLRP3 inflammasome in airway injury (PMID: 42211665).

Protein dysfunction / toxin (direct branch). The **CARDS toxin** is the key virulence factor — an **ADP-ribosylating and vacuolating** exotoxin that binds host membrane phospholipids (sphingomyelin, phosphatidylcholine) and annexin-A2 as receptors, recapitulating the inflammatory and histopathological damage seen in infected tissue (PMID: 41482890). Mp also generates **hydrogen peroxide**, causing direct oxidative cytotoxicity (PMID: 42211665).

Immune / autoimmune branch (molecular mimicry). Anti-galactocerebroside (anti-GalC) antibodies — GalC is a **major CNS myelin component** — were found in the CSF of **8/21 (31%)** Mp patients, implicating antibody-mediated demyelination (PMID: 17651816; PMID: 22419016). Anti-GM1/antiganglioside IgM antibodies accompany reversible basal-ganglia/thalamic involvement (PMID: 20305125). Short latency (**62% within two weeks**) supports a para-infectious immune mechanism (PMID: 41661491). ADEM occurs in ~12–18% of Mp neurologic cases (PMID: 26084846; PMID: 31549170).

Vaso-occlusive branch. Focal vasculitis, hypercoagulability, and hypoperfusion produce **cerebral watershed infarction**, sometimes with arterial thrombi and CSF MP-PCR positivity; elevated D-dimer marks the thrombotic/severe late-onset phenotype (PMID: 41398820; PMID: 41935250).

Upstream vs downstream. Upstream = respiratory colonization/infection. The three branches are parallel intermediate mechanisms. Downstream convergence = BBB disruption, neuroinflammation, edema, and neuronal dysfunction producing the clinical syndrome.

Suggested ontology terms: GO:0006954 (inflammatory response), GO:0002376 (immune system process), GO:0050900 (leukocyte migration), GO:0097190 (apoptotic signaling), GO:0006979 (response to oxidative stress); CL:0000129 (microglial cell), CL:0000127 (astrocyte), CL:0000540 (neuron); CHEBI:16240 (hydrogen peroxide), CHEBI:18390 (galactocerebroside class).

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7. Anatomical Structures Affected

Organ / system: Central nervous system (nervous system, UBERON:0001016); brain (UBERON:0000955). Primary respiratory-tract involvement (lung, UBERON:0002048) precedes/accompanies neurologic disease.

Regional CNS localization (by syndrome): - **Basal ganglia** (UBERON:0002420) and **thalamus** (UBERON:0001897) — symmetric T2/FLAIR hyperintensities in basal-ganglia encephalitis (PMID: 42475768; PMID: 20305125) - **Corpus callosum / splenium** (UBERON:0002336) — reversible splenial lesion (MERS) (PMID: 32307392) - **Cerebellum** (UBERON:0002037) — cerebellitis/ataxia (PMID: 26961264) - **Brainstem** (UBERON:0002298) — Bickerstaff brainstem encephalitis (PMID: 26115201) - **Hypothalamus** (UBERON:0001898) — secondary hypersomnia,

diabetes insipidus (PMID: 22419016) - **White matter / myelin** — demyelination (ADEM) - **Cerebral watershed territories** — infarction (PMID: 41398820) - **Leptomeninges** — leptomeningeal enhancement in 55.3% of adults (PMID: 42624009)

Tissue/cell level: Neurons (CL:0000540), oligodendrocytes/myelin (CL:0000128), microglia (CL:0000129), astrocytes (CL:0000127), and cerebral vascular endothelium.

Lateralization: Frequently **bilateral and symmetric** (basal ganglia, thalamus, splenium, cerebellum); infarcts may be focal/asymmetric.

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8. Temporal Development

Onset: Acute to subacute. Two temporal patterns partition the disease (see Key Findings): **early-onset (≤7 days from fever)** with direct/CSF-detectable invasion vs **late-onset (>7 days)** immune-mediated disease (PMID: 26084846; PMID: 41935250). Post-infectious autoimmune presentations show short latency, with 62% presenting within two weeks (PMID: 41661491).

Progression: Usually **monophasic**. Course ranges from mild, self-limited (MERS, transient basal-ganglia lesions that fully reverse) to fulminant with refractory status epilepticus. Reversible lesions may resolve within 1–4 weeks on imaging (PMID: 32307392; PMID: 20305125).

Remission / recovery: Most patients recover, many completely (e.g., 85.7% complete neurological recovery in the basal-ganglia subset at median 3.5 years; PMID: 42475768). **Critical intervention window:** early anti-mycoplasmal therapy plus timely immunomodulation during the acute phase.

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9. Inheritance and Population

Inheritance: Not applicable — infectious/multifactorial, no Mendelian inheritance, penetrance, anticipation, mosaicism, founder effect, or carrier frequency.

Epidemiology: - Neurologic disease is the most common extrapulmonary complication of Mp, incidence **~0.1% of infections** (PMID: 41697567). - *M. pneumoniae* accounts for **~5–6% of confirmed childhood encephalitis** and is the most frequently identified pathogen in several cohorts (see F001). - **Age:** pediatric predominance; MPE median ~7 years; adult

meningoencephalitis median ~40 years. - **Sex:** modest male predominance (43M:28F pediatric). - **Season/geography:** winter onset in ~half of cases; epidemics every 3–7 years; notable post-pandemic surge (2023–2024), with high macrolide resistance in China.

Cohort	N	Mp finding	PMID
California Encephalitis Project	1988 referred	111 acute Mp; 84 pediatric (76%); most frequent pathogen	17671056
Australian childhood encephalitis	287 confirmed	6% Mp	31549170
SickKids (16-yr)	42/365 PCR+	11.5% attributable neuro disease; encephalitis 52%	26084846

10. Diagnostics

Core principle: Diagnosis is **serology-driven** because direct CSF detection is insensitive. In the California Encephalitis Project, **80%** of Mp-associated cases were diagnosed by serology alone and **CSF PCR was positive in only 2%** (PMID: 17671056).

Laboratory / microbiology: - **Serology:** seroconversion, IgM positivity, or a single high titer (≥1:320); paired acute/convalescent titers preferred (PMID: 42624009). - **PCR:** respiratory-tract PCR often positive; **CSF PCR rarely positive** (marks the direct-invasive subset). - **Intrathecal antibody synthesis** can prove CNS infection (PMID: 26115201). - **Multiplex syndromic panels** (BioFire FA-ME, QIAstat-Dx) include Mp but detect it rarely in CSF, consistent with low CSF burden (PMID: 41716049). - **Inflammatory markers:** elevated CRP (90.8%); D-dimer and LDH mark severe/late-onset disease (PMID: 37417436; PMID: 42568601). - **CSF:** mild pleocytosis (~66.7% in basal-ganglia subset), oligoclonal bands (42.1%) (PMID: 42475768). - **Autoantibodies (subset):** anti-GalC, anti-GM1/ganglioside, anti-NMDAR.

Imaging: MRI abnormalities in ~79% of adults; patterns include symmetric basal-ganglia/thalamic T2/FLAIR hyperintensity, reversible splenial lesion (MERS), cerebellar involvement, white-matter (ADEM) lesions, and watershed infarcts. **EEG:** abnormal in ~36%; focal slow/delta waves predict adverse outcome.

Diagnostic categories: definite/probable/possible, graded by strength of microbiologic evidence (PMID: 26084846).

Differential diagnosis: viral encephalitis (HSV, enterovirus, HHV-6), **anti-NMDAR autoimmune encephalitis** (important overlap — see Section on Key Findings F008), ADEM of other cause, metabolic/toxic encephalopathy.

Genetic/omics testing: Not applicable for diagnosis (no causal gene). WGS/WES/panels/karyotyping/CMA have no role.

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11. Outcome / Prognosis

Overall: Generally favorable — most cases are acute/monophasic and recoverable — but severe forms carry high morbidity/mortality.

- **Adult meningoencephalitis:** 90-day mortality **2.6%** (unrelated to meningoencephalitis); neurologic sequelae **28.2%** (PMID: 42624009).
- **Pediatric MPE:** 9/71 poor outcome (GOS 3); late-onset associated with elevated D-dimer and severe pneumonia (PMID: 41935250).
- **Status epilepticus subset:** 67% developed refractory status epilepticus; at 18–86 months, 2 died, 7 developed epilepsy/deficits, **none returned to baseline** (PMID: 20682202).
- **Mycoplasma-associated acute encephalitis:** sequelae in 5/12 (42%) in one cohort (PMID: 17240177).
- **Reversible syndromes** (MERS, antibody-mediated basal-ganglia lesions) often fully reverse (PMID: 32307392; PMID: 20305125).

Prognostic factors for adverse outcome: focal neurological signs, multiple seizures or status epilepticus on admission, leukopenia, focal slow/continuous generalized delta waves on EEG, and focal cortical parenchymal hyperintensity on MRI (PMID: 17240177). Early-invasive disease generally carries worse outcomes than reversible immune-mediated forms.

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12. Treatment

Dual strategy: antibiotics + immunomodulation.

Anti-mycoplasmal antibiotics (must penetrate CNS and target a cell-wall–independent bacterium — **β-lactams are ineffective**): - **Macrolides** — azithromycin (NCIT: Azithromycin); limited by widespread A2063G resistance (PMID: 42568601) - **Tetracyclines** — doxycycline, **minocycline** (used successfully with clinical/CSF hypocretin recovery in an anti-GalC case; PMID: 22419016) - **Fluoroquinolones** — for macrolide-resistant strains

Immunomodulation (targets the autoimmune/inflammatory branches): - **Corticosteroids** (NCIT: Corticosteroid) — used in essentially all immune-mediated cases (PMID: 26961264; PMID: 20305125) - **IVIG** (intravenous immunoglobulin) - **Plasma exchange / plasmapheresis** — for severe/refractory or antibody-mediated disease - **Rituximab** — reported for anti-NMDAR overlap after steroids/IVIG/plasma exchange (PMID: 25222311)

Adjunctive: anticoagulation/thrombolysis (heparin, aspirin, urokinase) for the vaso-occlusive/infarction branch (PMID: 41398820); anticonvulsants for seizures/status epilepticus; supportive/rehabilitative care for sequelae.

Pharmacogenomics: Not established for MPE. **Personalized medicine** here means resistance-guided antibiotic selection (23S rRNA A2063G testing → non-macrolide agents) rather than host-genotype-guided therapy.

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13. Prevention

Primary prevention: No licensed *M. pneumoniae* vaccine exists. A **three-component subunit vaccine candidate** — P1 protein, P40/90 complex, and a detoxified CARDS toxin mutant — induced protective immunity in preclinical models (PMID: 42042806). General measures: respiratory hygiene, reducing transmission in crowded settings.

Secondary prevention: Early recognition of neurologic symptoms during/after Mp respiratory illness and prompt initiation of effective (resistance-appropriate) antibiotics plus immunotherapy to limit CNS injury.

Tertiary prevention: Aggressive seizure control to avert refractory status epilepticus; anticoagulation where thrombotic mechanisms operate; rehabilitation for sequelae.

Genetic counseling / screening: Not applicable (non-genetic disease). **Public-health surveillance** of epidemic cycles and macrolide-resistance genotypes supports population-level control (PMID: 42757852).

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14. Other Species / Natural Disease

Mycoplasma species cause **natural CNS disease across multiple animal species** (PMID: 28889017): - ***Mycoplasma agalactiae*** (sheep/goats, NCBI:txid2110) — found in large quantities in brain, causing **non-purulent encephalitis and ataxia** in young animals; CNS histopathology reproduced by experimental intra-mammary infection. - ***Mycoplasma bovis*** (cattle, NCBI:txid28903) — brain abscesses and fibrinous meningitis in calves/adult cattle. - **Avian *M. gallisepticum* / *M. synoviae*** — meningeal vasculitis and encephalitis in poultry. - **Marine mammals** — mycoplasmas isolated from brains of sea mammals. - **Spiroplasma** — implicated in transmissible spongiform encephalopathies.

Zoonotic potential: *M. pneumoniae* is a human-restricted pathogen; the animal mycoplasma encephalitides are largely species-specific and not zoonotic. **Comparative value:** these natural veterinary diseases demonstrate conserved mycoplasmal neuroinvasive and immune-mediated CNS injury mechanisms.

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15. Model Organisms

No animal model of *M. pneumoniae* extrapulmonary (neurologic) disease has been established (PMID: 41697567): *"the animal model of extrapulmonary manifestations caused by MP has not been established."* This is a major research gap. Existing Mp models focus on **respiratory** infection (mouse/hamster pneumonia models) and are used for vaccine and pathogenesis studies (e.g., CARDS toxin biology, subunit vaccine testing; PMID: 42042806). Naturally occurring animal mycoplasma encephalitides (Section 14) serve as **comparative** systems but do not model human MPE directly.

Key Findings (Expanded)

F001 — *M. pneumoniae* is a leading identifiable infectious cause of childhood encephalitis

Multiple independent cohorts converge on this conclusion. In the **California Encephalitis Project**, of 1988 referred patients, evidence of acute Mp infection was found in 111, of which 84 (76%) were pediatric, making Mp the most frequently identified pathogen (PMID: 17671056). A prospective Australian cohort of 287 confirmed childhood encephalitis cases placed Mp at **6%**, on par with HSV (PMID: 31549170). A 16-year Toronto SickKids series found 42/365 PCR-positive children (11.5%) with attributable neurologic disease, of which **encephalitis was 52%** (PMID: 26084846).

F002 — Two pathogenetic patterns: early-onset direct/invasive vs late-onset immune-mediated

The SickKids series identified two distinct patterns: (1) **prolonged prodrome (≥7 days)**, respiratory symptoms, a blood IgM response, and Mp detected in the respiratory tract **but not CSF** (immune-mediated); and (2) **brief (<7 days) or no prodrome** with Mp detected in CSF (direct invasion) (PMID: 26084846). This early/late split was reproduced in a pediatric MPE cohort (PMID: 41935250). Anti-GalC autoantibodies in CSF of 8/21 (31%) supports the autoimmune branch (PMID: 17651816).

F003 — Clinical phenotype: altered consciousness, seizures, movement disorders; frequent sequelae, low mortality

Pediatric MPE (n=87): consciousness disturbance 90%, headache 87.4%, fever 96.5%, respiratory involvement 94.3%, elevated CRP 90.8% (PMID: 37417436). Adult meningoencephalitis (n=39): 90-day mortality 2.6% (unrelated), sequelae 28.2% (PMID: 42624009).

F004 — Diagnosis relies on serology/PCR because CSF Mp PCR is rarely positive

80% of Mp-associated cases were diagnosed by serology alone; CSF PCR positive in only 2% (PMID: 17671056). Intrathecal antibody synthesis can prove CNS infection (PMID: 26115201).

F005 — Three mechanistic branches of CNS injury

Direct/toxin-mediated (CARDS toxin, MyD88/IL-8; PMID: 41482890, PMID: 42572866); autoimmune/molecular mimicry (anti-GalC, anti-GM1, short latency; PMID: 17651816, PMID: 41661491); vaso-occlusive/thrombotic (watershed infarction, elevated D-dimer; PMID: 41398820).

F006 — Broad CNS syndrome spectrum; encephalitis most common

Neurologic disease is the most common extrapulmonary complication (~0.1%; PMID: 41697567). Syndromes: encephalitis (52%), ADEM (12%), transverse myelitis (12%), cerebellar ataxia (10%) (PMID: 26084846), plus basal-ganglia encephalitis, MERS, Bickerstaff brainstem encephalitis, opsoclonus-myoclonus-ataxia, and stroke.

F007 — Epidemiology and treatment

Pediatric predominance (median 7 y, 43M:28F, winter onset in half; PMID: 41935250); macrolide resistance via A2063G (PMID: 42568601); dual antibiotic + immunomodulatory therapy; preclinical subunit vaccine candidate (PMID: 42042806).

F008 — Overlap with anti-NMDAR autoimmune encephalitis

In the California Encephalitis Project, 4 of 8 non-teratoma anti-NMDAR encephalitis patients had serologic evidence of acute Mycoplasma infection (PMID: 19718525). A pediatric anti-NMDAR case with Mp IgM was treated with steroids/IVIG/plasma exchange/rituximab (PMID: 25222311). Immune-mediated encephalitis affects older children than infectious encephalitis (median 7.6 vs 1.7 y; PMID: 31549170).

F009 — Comparative/veterinary disease and absence of an MPE animal model

Mycoplasmas cause natural CNS disease across species (PMID: 28889017); no animal model of human Mp extrapulmonary disease exists (PMID: 41697567).

F010 — Host genetic/immune-maturity factors modulate severity

Candidate susceptibility loci (ACE, IL-6, NOS3, IL2, IL4) for Mp pneumonia (PMID: 28403117); identical resistant strains produced divergent intrafamilial severity, implicating host immune maturity over virulence (PMID: 19898978).

F011 — Prognosis: mostly recoverable, but refractory status epilepticus carries high morbidity/mortality

MPE with status epilepticus: 67% refractory, 2 died / 7 with epilepsy or deficits, none returned to baseline (PMID: 20682202). Validated adverse-outcome predictors (PMID: 17240177).

Mechanistic Model / Interpretation

MPE is best understood as a **single trigger (M. pneumoniae infection) producing three convergent injury pathways**, partitioned in time by the length of the respiratory prodrome:

Feature	Early-onset (direct/ invasive)	Late-onset (immune-mediated)	Vaso-occlusive
Prodrome	Brief (<7 d) or none	Prolonged (≥7 d)	Variable
Organism in CSF	Yes (PCR+)	No (respiratory only)	Sometimes
Dominant mechanism	CARDS toxin, H_2O_2, TLR–MyD88–IL-8, NLRP3	Molecular mimicry (anti-GalC, anti-GM1, anti-NMDAR) → ADEM	Vasculitis, hypercoagulability, hypoperfusion
Peripheral marker	Direct detection	Blood IgM response	Elevated D-dimer
Representative syndrome	Fulminant encephalitis, status epilepticus	ADEM, basal-ganglia encephalitis, cerebellitis	Watershed infarction
Reversibility	Often worse	Frequently reversible	Variable
Key evidence	26084846, 41482890, 42572866	17651816, 20305125, 41661491	41398820, 41935250

This tripartite model explains why MPE demands **both** anti-mycoplasmal antibiotics (for the invasive branch) **and** immunomodulation (for the autoimmune branch), sometimes with anticoagulation (for the thrombotic branch) — and why host immune maturity, not organism virulence, dominates severity.

Evidence Base

PMID	Contribution
17671056	Mp = most frequent identified pathogen in California Encephalitis Project; serology-driven diagnosis, CSF PCR+ only 2%
31549170	Mp = 6% of confirmed childhood encephalitis; infectious younger than immune-mediated
26084846	Two pathogenetic patterns by prodrome; syndrome spectrum (encephalitis 52%)
37417436	Quantitative pediatric MPE phenotype
42624009	Adult meningoencephalitis: imaging, 2.6% mortality, 28.2% sequelae
41482890	CARDS toxin — ADP-ribosylating/vacuolating virulence factor, phospholipid receptor
42572866	CARDS TX/MyD88/IL-8 correlate with severity
17651816	Anti-GalC in CSF of 31% of Mp patients (autoimmune branch)
41661491	Short latency (62% <2 weeks) supports para-infectious immunity
19718525	Mp–anti-NMDAR link and differential diagnosis
28889017	Comparative/veterinary mycoplasma encephalitis
41697567	Neuro = most common extrapulmonary complication (~0.1%); no animal model
19898978	Host immune maturity > virulence for severity
20682202	High morbidity of MPE with status epilepticus
17240177	Validated prognostic predictors
42568601	A2063G macrolide resistance; D-dimer/LDH severity predictors
42042806	Preclinical three-component subunit vaccine (P1, P40/90, detoxified CARDS)

Limitations and Knowledge Gaps

1. **Definition and coding:** MPE lacks a dedicated OMIM/Orphanet entry and specific ICD code; it is captured via combinations of encephalitis + Mp infection codes, limiting registry-based epidemiology.
2. **Diagnostic uncertainty:** Because CSF PCR is positive in only ~2%, most diagnoses rest on serology, which cannot always distinguish causation from coincident infection — the watershed-infarction case series explicitly asked whether the association was coincidental (PMID: 41398820).
3. **No animal model** of extrapulmonary/neurologic disease impedes mechanistic dissection and therapeutic testing (PMID: 41697567).
4. **Autoimmune mechanism** evidence is largely from case series and small cohorts; anti-GalC and antiganglioside antibodies are found in only a subset, and the causal weight of each branch per patient is often uncertain.
5. **Host genetics** data pertain to respiratory disease susceptibility (mostly Chinese pediatric cohorts) and may not generalize or apply to encephalitis specifically.
6. **Treatment evidence** is observational; no randomized trials define optimal antibiotic/immunotherapy combinations or timing for MPE.

Proposed Follow-up Experiments / Actions

1. **Develop a murine neuroinvasion model** using intranasal Mp challenge with CARDS-toxin-competent vs detoxified strains to test the direct-injury branch and evaluate CNS-penetrating therapeutics.
2. **Prospective biomarker stratification:** Combine CSF PCR, paired serology, anti-GalC/anti-GM1/anti-NMDAR panels, D-dimer, and cytokines (IL-8, MyD88-axis) to prospectively classify patients into the three mechanistic branches and match therapy accordingly.
3. **Randomized/adaptive trial** of antibiotic-plus-immunotherapy timing (early corticosteroids/IVIG vs standard) stratified by early- vs late-onset phenotype.
4. **Resistance-guided treatment study:** Evaluate whether rapid 23S rRNA A2063G genotyping and non-macrolide selection improves neurologic outcomes during high-resistance epidemics.

5. **Standardize MPE case definitions and ontology mapping** (MONDO, HPO, UBERON) to enable pooled multicenter registries and improve epidemiologic precision.
6. **Advance CARDS-toxin subunit vaccine** (PMID: 42042806) and assess whether toxin neutralization reduces neurologic complications.

Evidence source types: predominantly human clinical (cohorts, case series, systematic reviews); mechanistic/in vitro (CARDS toxin biology); comparative/veterinary (natural animal mycoplasma encephalitis); no established model-organism data for the neurologic disease.
